

Exact Single-Monomial High-Order Partial Derivatives via Waring Theory and Taylor Directional Derivatives

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Abstract

We present a deterministic Taylor-jet backend for evaluating one fixed high-order mixed partial derivative of a neural-network surrogate, a workload that appears repeatedly in scientific machine learning and high-order physics-informed neural networks. The central observation is that a single multi-index derivative is a coefficient of a directional Taylor polynomial; hence the problem of designing a shortest exact set of directional probes is equivalent to a Waring decomposition of the corresponding monomial. The Carlini–Catalisano–Geramita theorem gives the minimum number of complex directions in closed form and an explicit roots-of-unity construction. We combine this schedule with Taylor-mode propagation through multilayer perceptrons, using the entire activation $\sinh$ when complex parameters are used, to obtain an exact floating-point implementation that is differentiable with respect to network parameters. Numerical experiments on direct derivative tasks and a four-dimensional manufactured sixth-order PDE compare the method with nested automatic differentiation and real polarization. The results confirm round-off-level agreement with the direct reference and show that, for repeated-index multi-indices, the Waring schedule reduces the number of Taylor directions while remaining competitive in measured wall-clock time. Because $\sinh$ is entire on $\mathbb{C}$, the same backend also makes *complex-parameter* $\sinh$ networks high-order differentiable without any runtime dtype conversion; we study these as a PINN ansatz for oscillatory and high-order problems and benchmark them, under matched parameter counts and a shared training protocol, against strong real high-frequency baselines (Fourier features, SIREN, MscaleDNN) on a literature-grounded suite of nine PDE families. The complex network is competitive-to-superior on the majority of instances, with the largest gains on balanced repeated-index high-order operators and on genuinely complex-valued problems.

Keywords. Physics-informed neural networks; high-order automatic differentiation; Waring decomposition; apolarity; Taylor-mode AD; single-monomial partial derivative.

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1 Introduction

1.1 The problem

Many scientific machine-learning workflows ask, repeatedly during training, for the value of a single fixed high-order mixed partial derivative of a neural-network surrogate. Examples include physics-informed neural networks (PINNs) [14] for high-order PDEs (Korteweg–de Vries, Cahn–Hilliard, biharmonic, polyharmonic), neural-network variational Monte Carlo [13, 6], and stochastic optimal control via the Hamilton–Jacobi–Bellman equation.

The naive evaluation chains successive reverse-mode differentiations through the network. The resulting autograd graph deepens linearly with both the differentiation order and the network depth, and the wall-clock cost becomes the bottleneck of training as soon as the order moves beyond two or three. Three classes of acceleration techniques have been proposed:

- (i) **Forward-mode / Taylor-mode automatic differentiation** [1, 4], which propagates truncated Taylor coefficients through every primitive of the network in a single forward pass and avoids the autograd-graph chaining overhead.
- (ii) **Stochastic estimators** [7, 15], which write the target operator as an expectation over random tangent vectors and estimate it by Monte Carlo sampling.
- (iii) **Operator-specific specializations** [11, 3] for the Laplacian (the trace of the Hessian), exploiting the fact that the trace collapses cleanly through linear layers.

Each has limitations. Forward-mode AD by itself does not say how to combine information from several directions. Stochastic estimators introduce a variance that depends on the sampling budget. And operator-specific methods do not extend to non-trace multi-indices.

This paper develops a fourth, complementary route: an exact deterministic backend for one fixed multi-index at a time, based on identifying the directional schedule problem with a classical combinatorial-algebraic problem – the Waring decomposition of a monomial.

1.2 Contributions

1. **Bridge theorem (Section 3).** We prove that a directional schedule – a finite linear combination of directional Taylor coefficients reproducing the target derivative for every smooth input – exists with a given direction count if and only if a single monomial of degree equal to the differentiation order admits a Waring decomposition of that length. Consequently, the smallest possible direction count equals the Waring rank of the corresponding monomial.
2. **Closed-form schedule (Section 3).** The Carlini–Catalisano–Geramita rank theorem [2] and its roots-of-unity construction yield, for any fixed multi-index, an explicit complex schedule whose direction count attains the information-theoretic minimum over the complex field. For some repeated-index patterns this strictly improves on classical real polarization; for square-free patterns the direction counts coincide.
3. **End-to-end algorithm (Section 4).** We pair the schedule above with a forward-mode Taylor-jet propagation through sinh-activated multilayer perceptrons. Because sinh is entire on the complex plane, the jet rules are well-defined on complex inputs and complex parameters, and the resulting algorithm is exact up to floating-point round-off.
4. **Scientific-computing evaluation (Section 5).** We benchmark direct derivative queries and a four-dimensional manufactured sixth-order PDE. The comparison uses the two implemented deterministic baselines – direct nested automatic differentiation and real polarization with Taylor jets – and reports both direction counts and measured wall-clock costs. Within a fixed wall-clock budget, the complex Waring backend gives the best result in the single tested PDE run; we report this as fixed-budget evidence rather than a statistical superiority claim.
5. **Oscillatory expressivity study (Sections 6–7).** The entirety of sinh on $\mathbb{C}$ lets the same backend differentiate *complex-parameter* sinh networks to arbitrary order with no dtype casting in the inner loop. We treat such a network as a PINN ansatz and ask whether the complex value field is itself more expressive on oscillatory and high-order targets. On a literature-grounded suite of nine PDE families – evaluated under matched real-parameter counts, matched frequency initialization, and one shared training protocol – the complex sinh network is competitive-to-superior against Fourier-feature, SIREN, and MscaledNN baselines on most instances, with the clearest advantage on balanced repeated-index high-order operators (e.g. Δ^3 , mixed plate modes) and on genuinely complex-valued problems (nonlinear Schrödinger, lossy Maxwell). We report the advantage and its limits honestly, including the families where the gap closes.

1.3 Scope and limitations

The *rank advantage* of this work is specific to one fixed multi-index: it shortens the directional schedule of a single repeated-index mixed partial. Operator sums such as the Laplacian or the polyharmonic operator Δ^m are still evaluated exactly by the same jet machinery – each summand is one (rank-one) directional jet and the residual is their sum – but for such sums the Waring rank offers no shortening, and dedicated trace-collapsing techniques [11, 3] can be more efficient when only a Laplacian is needed. The oscillatory study of Sections 6–7 uses the backend in this summed, exact-but-not-shortened mode wherever the operator is a sum, and exploits the genuine rank advantage on the mixed high-order terms. We also do not pursue stochastic estimation; its accuracy-cost tradeoffs are complementary to ours [15].

1.4 Relation to existing approaches

The method sits between general automatic differentiation and operator-specific scientific-computing accelerators. Taylor-mode AD [1, 4, 18] supplies the univariate directional derivatives, but it does not determine which finite set of directions extracts a prescribed mixed partial. Real polarization provides a deterministic schedule with real directions, at the cost of up to 2^{p-1} coalesced probes. Stochastic derivative estimators [7, 15] address broader tensor contractions through sampling and are therefore complementary rather than direct replacements for an exact single-monomial schedule. Trace-specialized methods for Laplacians [11, 3] exploit operator sums and are outside the single-monomial setting. Our contribution is to use the monomial Waring rank to make the deterministic direction schedule minimal over $\mathbb{C}$ for the fixed multi-index problem.

1.5 Notation

Throughout the paper, $u : \mathbb{R}^d \rightarrow \mathbb{R}$ denotes a sufficiently smooth scalar field. A multi-index of order p is an ordered tuple $\alpha = (i_1, \dots, i_p) \in \{1, \dots, d\}^p$, encoding the operator

$$\partial^\alpha := \partial_{i_1} \partial_{i_2} \cdots \partial_{i_p}. \quad (1)$$

The *active coordinates* of α are its distinct entries, relabelled as $i_0, \dots, i_n$ so that their multiplicities $a_0, \dots, a_n$ satisfy $1 \leq a_0 \leq \dots \leq a_n$ (ties may be broken by the original coordinate index). This ordering is not, in general, the coordinate ordering; it singles out a minimum-multiplicity variable i_0 , as required by the monomial Waring rank formula below. We set

$$p = \sum_{j=0}^n a_j, \quad \alpha! = \prod_{j=0}^n a_j!, \quad m_j = a_j + 1. \quad (2)$$

The *active subspace* V is the linear span of the active coordinate directions,

$$V := \text{span}\{e_{i_0}, \dots, e_{i_n}\} \subset \mathbb{R}^d; \quad (3)$$

since ∂^α is insensitive to coordinates outside the active set, all directional probes v below may be taken in V .

The directional Taylor coefficient of order p at x along $v \in V$ is

$$T_p(x; v) := \frac{1}{p!} \frac{d^p}{d\tau^p} u(x + \tau v) \Big|_{\tau=0} = \frac{1}{p!} D^p u(x)[v, \dots, v]. \quad (4)$$

Throughout the paper, the default scalar field is the real field $\mathbb{R}$; we write $\mathbb{R}[z]_p$ for the space of homogeneous polynomials of degree p in $n+1$ variables $z = (z_0, \dots, z_n)$ over $\mathbb{R}$. The complex extension $\mathbb{C}[z]_p$ is needed only in Section 3.2, where the closed-form schedule of Theorem 3.6 uses complex coefficients and complex linear forms; we introduce it explicitly there.

2 Single-Monomial Partial as Polynomial Coefficient Extraction

2.1 The generating polynomial

For $z = (z_0, \dots, z_n) \in \mathbb{R}^{n+1}$ parametrizing the active subspace via $v = \sum_j z_j e_{i_j}$, the *generating polynomial* of u at x is

$$Q_u(z) := T_p\left(x; \sum_{j=0}^n z_j e_{i_j}\right), \quad z \in \mathbb{R}^{n+1}. \quad (5)$$

By construction Q_u is a degree- p homogeneous polynomial in z ; the next lemma identifies its coefficients with the partial derivatives of u .

Lemma 2.1 (Generating polynomial expansion). *For $u \in C^p$ on a neighborhood of x ,*

$$Q_u(z) = \sum_{|\beta|=p} \frac{\partial^\beta u(x)}{\beta!} z^\beta, \quad (6)$$

where β ranges over multi-indices on the active variables and $\partial^\beta = \partial_{i_0}^{\beta_0} \dots \partial_{i_n}^{\beta_n}$.

Proof. By the symmetric multilinearity of $D^p u(x)$ and the multinomial theorem,

$$D^p u(x) \left[\sum_j z_j e_{i_j}, \dots, \sum_j z_j e_{i_j} \right] = \sum_{|\beta|=p} \frac{p!}{\beta!} z^\beta \partial^\beta u(x).$$

Dividing by $p!$ yields (6). □

Corollary 2.2. $\partial^\alpha u(x) = \alpha! [z^\alpha] Q_u(z)$, where $[z^\alpha]$ extracts the coefficient of the monomial z^α .

2.2 Directional schedules

A *directional schedule* of length R for ∂^α is a finite collection $(c_r, v_r)_{r=1}^R \subset \mathbb{R} \times V$ such that

$$\partial^\alpha u(x) = \sum_{r=1}^R c_r T_p(x; v_r), \quad \forall u \in C_x^p. \quad (7)$$

Because the schedule is required to hold for every $u \in C_x^p$, the p -jet of u at x may be prescribed arbitrarily on the active variables: for any coefficients q_β , the polynomial $\sum_\beta q_\beta \prod_j (y_{i_j} - x_{i_j})^{\beta_j}$ has $\partial^\beta u(x) = \beta! q_\beta$. This observation lets us test identities coefficient by coefficient in $\mathbb{R}[z]_p$.

The central design question of this paper is then: *What is the minimum length R^* of a directional schedule (7), and how is one constructed?* Section 3 reduces this to a classical algebraic question.

3 Equivalence to Monomial Waring Decomposition

3.1 The bridge theorem

For $v \in V$, write $v = \sum_{j=0}^n v_j e_{i_j}$ in active coordinates. We pair this direction with the polynomial variable $z = (z_0, \dots, z_n)$ by $v \cdot z = \sum_{j=0}^n v_j z_j$.

Theorem 3.1 (Bridge). *Let $R \in \mathbb{N}$, $c_r \in \mathbb{R}$, $v_r \in V$, $r = 1, \dots, R$. The following are equivalent.*

- (1) (*Directional schedule*) $\partial^\alpha u(x) = \sum_{r=1}^R c_r T_p(x; v_r) \quad \forall u \in C_x^p$.

(2) (*Polynomial identity in $\mathbb{R}[z]_p$*) $p! z^\alpha = \sum_{r=1}^R c_r (v_r \cdot z)^p$.

Proof. Write $v_r = (v_{r,0}, \dots, v_{r,n})$ in active coordinates. By Lemma 2.1,

$$T_p(x; v_r) = \sum_{|\beta|=p} \frac{\partial^\beta u(x)}{\beta!} v_r^\beta.$$

Substituting this expansion into statement (1), and using that the order- p active jet of u at x can be prescribed arbitrarily, shows that (1) is equivalent to the finite-dimensional coefficient system

$$\sum_{r=1}^R c_r v_r^\beta = \alpha! \delta_{\alpha\beta}, \quad \forall \beta \in \mathbb{Z}_{\geq 0}^{n+1} \text{ with } |\beta| = p. \quad (\star)$$

On the other hand, expanding statement (2) gives $(v_r \cdot z)^p = \sum_{|\beta|=p} \binom{p}{\beta} v_r^\beta z^\beta$ and hence, after comparing coefficients of z^β ,

$$\binom{p}{\beta} \sum_{r=1}^R c_r v_r^\beta = p! \delta_{\alpha\beta}, \quad \forall |\beta| = p.$$

Using $p!/\binom{p}{\beta} = \beta!$ and $\beta! \delta_{\alpha\beta} = \alpha! \delta_{\alpha\beta}$, this is exactly $(\star)$. Thus (1) and (2) are equivalent. $\square$

Definition 3.2 (Waring rank). For a homogeneous polynomial $F \in \mathbb{R}[z]_p$, its *real Waring rank* is

$$R_{\mathbb{R}}(F) := \min \left\{ R : F = \sum_{r=1}^R c_r L_r(z)^p, \ c_r \in \mathbb{R}, \ L_r \in \mathbb{R}[z]_1 \right\}.$$

The *complex Waring rank* $R_{\mathbb{C}}(F)$ is defined identically with $\mathbb{R}$ replaced by $\mathbb{C}$; since $\mathbb{R} \subset \mathbb{C}$, $R_{\mathbb{C}}(F) \leq R_{\mathbb{R}}(F)$.

Corollary 3.3. *The minimum length R^* of a directional schedule for ∂^α equals the real Waring rank $R_{\mathbb{R}}(z^\alpha)$ of the monomial.*

Remark 3.4 (Complex extension). Theorem 3.1 and its proof go through verbatim if the coefficients c_r and the directions v_r are allowed to take values in $\mathbb{C}$ and $\mathbb{C}^{n+1}$, and the polynomial identity is read in $\mathbb{C}[z]_p$. The minimum length of such a complex schedule then equals $R_{\mathbb{C}}(z^\alpha)$. The next subsection gives this complex rank in closed form.

3.2 The Waring rank of a monomial

The closed-form rank over $\mathbb{C}$ is classical. (Problem (P)/(Q) is posed over $\mathbb{R}$ by default; passing to $\mathbb{C}$ can only decrease the rank, so $R_{\mathbb{C}} \leq R_{\mathbb{R}}$.)

Theorem 3.5 (Carlini–Catalisano–Geramita [2]). *For $z^\alpha = z_0^{a_0} \cdots z_n^{a_n}$ with $1 \leq a_0 \leq \cdots \leq a_n$,*

$$R_{\mathbb{C}}(z^\alpha) = \prod_{j=1}^n (a_j + 1) = \frac{\prod_{j=0}^n (a_j + 1)}{a_0 + 1}. \quad (8)$$

The lower bound, sketched in [2], uses the apolar ideal $I_{z^\alpha} = (\partial_0^{a_0+1}, \dots, \partial_n^{a_n+1})$ (a complete intersection); we recall the argument in Appendix A.3. The upper bound is constructive.

Theorem 3.6 (Roots-of-unity directional schedule). *Let $m_j = a_j + 1$ for $j = 1, \dots, n$, and for each tuple $\zeta = (\zeta_1, \dots, \zeta_n)$ with $\zeta_j^{m_j} = 1$ define*

$$v_\zeta = e_{i_0} + \sum_{j=1}^n \zeta_j e_{i_j}, \quad c_\zeta = \frac{\alpha!}{\prod_{j=1}^n m_j} \prod_{j=1}^n \zeta_j. \quad (9)$$

Then $\partial^\alpha u(x) = \sum_\zeta c_\zeta T_p(x; v_\zeta)$ for every $u \in C_x^p$. The schedule has length $\prod_{j=1}^n m_j = R_{\mathbb{C}}(z^\alpha)$ and is therefore length-optimal over $\mathbb{C}$.

The proof, given in Appendix A.2, verifies condition (2) of Theorem 3.1 directly via the roots-of-unity filter $\sum_{\zeta^m=1} \zeta^k = m \mathbf{1}_{m|k}$.

To illustrate how Theorem 3.1 couples the analytic side (7) (problem (P)) with the algebraic side $p! z^\alpha = \sum_r c_r (v_r \cdot z)^p$ (problem (Q)), we work out two examples: a low-order one for which both sides reduce to hand-checkable identities, and a higher-order repeated-index one that exhibits the full strength of the complex Waring construction.

Example 3.7 ($p = 2$, $\alpha = (1, 1)$, real schedule of length 2). For $\alpha = (1, 1)$ we have $n = 1$, $a_0 = a_1 = 1$, $m_1 = 2$, $U_2 = \{+1, -1\}$. Equation (9) produces the schedule

$$(c_{+1}, v_{+1}) = (\tfrac{1}{2}, e_1 + e_2), \quad (c_{-1}, v_{-1}) = (-\tfrac{1}{2}, e_1 - e_2). \quad (10)$$

Verification on the analytic side. Expanding the directional Taylor coefficient $T_p(x; v) = \frac{1}{2}(u_{11}v_1^2 + 2u_{12}v_1v_2 + u_{22}v_2^2)$,

$$\tfrac{1}{2}T_p(x; e_1 + e_2) - \tfrac{1}{2}T_p(x; e_1 - e_2) = \tfrac{1}{4}[(u_{11} + 2u_{12} + u_{22}) - (u_{11} - 2u_{12} + u_{22})] = u_{12}(x).$$

Verification on the algebraic side. The equivalent polynomial identity in $\mathbb{R}[z]_2$ reads

$$2! z_0 z_1 = \tfrac{1}{2}(z_0 + z_1)^2 - \tfrac{1}{2}(z_0 - z_1)^2 = \tfrac{1}{2}[(z_0^2 + 2z_0 z_1 + z_1^2) - (z_0^2 - 2z_0 z_1 + z_1^2)] = 2 z_0 z_1,$$

matching $p! z^\alpha = 2 z_0 z_1$. The two checks coincide via the bridge of Theorem 3.1.

Example 3.8 ($p = 6$, $\alpha = (2, 2, 2)$, complex schedule of length 9). Consider $u_{112233} = \partial_1^2 \partial_2^2 \partial_3^2 u$, i.e. $\alpha = (2, 2, 2)$. Here $n = 2$, $a_0 = a_1 = a_2 = 2$, $m_1 = m_2 = 3$; take $\omega = e^{2\pi i/3}$ and $U_3 = \{1, \omega, \omega^2\}$. Equation (9) produces $R_{\mathbb{C}}(z^{(2,2,2)}) = 9$ directions: for each $(\zeta_1, \zeta_2) \in U_3 \times U_3$,

$$v_\zeta = e_1 + \zeta_1 e_2 + \zeta_2 e_3, \quad c_\zeta = \frac{(2!)^3}{3 \cdot 3} \zeta_1 \zeta_2 = \frac{8}{9} \zeta_1 \zeta_2. \quad (11)$$

Analytic side. The directional schedule reads

$$u_{112233}(x) = \frac{8}{9} \sum_{\zeta_1, \zeta_2 \in U_3} \zeta_1 \zeta_2 T_p(x; e_1 + \zeta_1 e_2 + \zeta_2 e_3). \quad (12)$$

Algebraic side. The equivalent polynomial identity in $\mathbb{C}[z]_6$ is

$$6! z_0^2 z_1^2 z_2^2 = \frac{8}{9} \sum_{\zeta_1, \zeta_2 \in U_3} \zeta_1 \zeta_2 (z_0 + \zeta_1 z_1 + \zeta_2 z_2)^6. \quad (13)$$

That (13) holds is verified by the roots-of-unity filter (Lemma A.1): expanding the right-hand side multinomially and collecting the coefficient of $z_0^{\beta_0} z_1^{\beta_1} z_2^{\beta_2}$ reduces to the product $\prod_{j=1}^2 \sum_{\zeta_j \in U_3} \zeta_j^{1+\beta_j}$, which is non-zero exactly when $\beta_j \equiv 2 \pmod{3}$ for $j = 1, 2$; together with $\sum_j \beta_j = 6$ this forces $\beta = (2, 2, 2)$, and the leading constant $\frac{8}{9} \cdot \frac{6!}{2!2!2!} \cdot 9 = 720 = 6!$ gives the right normalisation.

By comparison, the coalesced antipodally merged real polarization schedule used in our baseline requires 13 distinct nonzero directions for the same $u_{112233}(x)$; the complex Waring schedule above attains the information-theoretic complex lower bound $R_{\mathbb{C}} = 9$ (Theorem 3.5).

3.3 Pattern taxonomy

Table 1 lists the rank values for representative active-exponent patterns up to $p = 6$, comparing the complex Waring rank $R_{\mathbb{C}}$ of Theorem 3.5 with the post-antipodal polarization count 2^{p-1} , the latter being the direction count attainable by classical real polarization for a generic multi-index of order p .

pattern	example multi-index	R_C	polarization
(p)	$\partial_1^p u$	1	$\leq 2^{p-1}$
$(2, 1)$	u_{112}	3	3
$(1, 1, 1)$	u_{123}	4	4
$(3, 1)$	u_{1112}	4	4
$(2, 2)$	u_{1122}	3	4
$(2, 1, 1)$	u_{1123}	6	6
$(1, 1, 1, 1)$	u_{1234}	8	8
$(4, 2)$	u_{111122}	5	7
$(2, 2, 2)$	u_{112233}	9	13
(1^6)	u_{123456}	32	32

Table 1: Bold entries mark patterns where the complex Waring schedule is strictly shorter than the polarization count. The square-free pattern (1^p) saturates the lower bound 2^{p-1} ; the method of this paper offers no improvement there.

4 Taylor-Mode Evaluation and Backend Dispatch

4.1 The Taylor jet for Linear-sinh MLPs

Given a single direction v , Theorem 3.6 reduces the problem to evaluating $T_p(x; v)$. We use forward-mode Taylor propagation [1, 4], replacing each intermediate activation $y \in \mathbb{R}^B$ by a length- $(p+1)$ tuple $(y_0, \dots, y_p)$ representing its truncated Taylor coefficients $y_k = (1/k!) d^k y / d\tau^k|_{\tau=0}$.

For an MLP $u = \phi_L \circ A_L \circ \dots \circ \phi_1 \circ A_1$ with affine A_ℓ and entrywise activations ϕ_ℓ , the jet rules are as follows. The affine layer $y = Wx + b$ propagates each jet order independently:

$$y_0 = Wx_0 + b, \quad y_k = Wx_k \quad (k \geq 1). \quad (14)$$

The tanh activation, with auxiliary jet $z = 1 - y^2$, satisfies

$$y_k = \frac{1}{k} \sum_{j=0}^{k-1} (k-j) z_j x_{k-j}, \quad z_k = - \sum_{j=0}^k y_j y_{k-j}, \quad k \geq 1. \quad (15)$$

The sinh activation, with auxiliary jet $z = \cosh(x)$, satisfies the coupled recurrence

$$y_k = \frac{1}{k} \sum_{j=0}^{k-1} (k-j) z_j x_{k-j}, \quad z_k = \frac{1}{k} \sum_{j=0}^{k-1} (k-j) y_j x_{k-j}, \quad k \geq 1. \quad (16)$$

The sinh recurrence is identical to that of $\sin/\cos$ but without the sign flip. These recurrences are algebraic identities and can be evaluated on complex tensors. For the global complex-direction backend used in the PDE experiment we use $\sinh$, because $\sinh$ and $\cosh$ are entire on $\mathbb{C}$. The tanh rule is kept for real-valued experiments and local complex evaluations, but tanh is meromorphic and has poles in the complex plane.

Each Taylor coefficient y_k is a regular tensor, and the jet propagation has no autograd-graph chaining, so storage is $(p+1)$ times the ordinary forward-pass activation size. We additionally use a custom `torch.autograd.Function` for each activation jet, with the closed-form backward

$$g_x^{(m)} = \sum_{j=0}^{p-m} \bar{q}_j g_y^{(m+j)}, \quad q = \phi', \quad (17)$$

which avoids reverse differentiation through the intra-jet recurrence and is essential for memory-efficient training.

4.2 Backends

We compare three implemented deterministic backends for evaluating one fixed $\partial^\alpha u(x)$. All three are differentiable in the network parameters θ , so each can be used directly as the residual term of a PINN loss. The implementation also provides an `auto` dispatch option, but it is a selection rule between B2 and B3 rather than a separate mathematical backend.

(B1) Direct nested AD. The reference baseline performs p successive calls to PyTorch’s `autograd.grad`, each with retained computation graph. This induces a reverse-mode autograd graph of depth $O(pL)$ for a depth- L network.

(B2) Real polarization with Taylor jet. The classical antipodally merged polarization identity is

$$\partial^\alpha u(x) = \frac{1}{2^p} \sum_{\epsilon \in \{\pm 1\}^p} \left(\prod_{r=1}^p \epsilon_r \right) T_p \left(x; \sum_{r=1}^p \epsilon_r e_{i_r} \right). \quad (18)$$

After antipodal merging this yields at most 2^{p-1} real directions. The directional Taylor coefficients $T_p(x; \cdot)$ are evaluated by the jet propagation of Section 4.

(B3) Complex Waring with Taylor jet. The roots-of-unity construction of Theorem 3.6 produces $R_{\mathbb{C}}(z^\alpha) = \prod_{j=1}^n (a_j + 1)$ directions in $\mathbb{C}^{n+1}$; for several repeated-index patterns this is smaller than the coalesced polarization count of (B2). For real-parameter networks, the complex directions induce per-call casts of the linear weights via the differentiable `Tensor.to` operator; for complex-parameter networks (Section 5.4) no cast is required and the network itself acts on the complex directions natively.

4.3 The end-to-end algorithm

Algorithm 1 summarises the complete pipeline of backend (B3): given a smooth model u (an MLP whose primitives are covered by the jet rules (14)–(16)), a point x , and a multi-index α of order p , it returns the value $\partial^\alpha u(x)$ without any nested back-propagation. Backend (B2) follows the same overall structure with the Waring schedule (lines 3–7) replaced by the polarization schedule (18).

Algorithm 1 Single-monomial $\partial^\alpha u(x)$ via complex Waring schedule + Taylor jet.

Require: smooth scalar field u ; point $x \in \mathbb{R}^d$; multi-index α of order p

Ensure: the value $\partial^\alpha u(x)$

- 1: relabel the active coordinates of α as $i_0, \dots, i_n$ with multiplicities $1 \leq a_0 \leq \dots \leq a_n$
 - 2: $m_j \leftarrow a_j + 1$ for $j = 1, \dots, n$; set $\mathcal{I} \leftarrow U_{m_1} \times \dots \times U_{m_n}$
 - 3: initialise schedule $\mathcal{S} \leftarrow \emptyset$
 - 4: **for all** $\zeta = (\zeta_1, \dots, \zeta_n) \in \mathcal{I}$ **do**
 - 5: $v_\zeta \leftarrow e_{i_0} + \sum_{j=1}^n \zeta_j e_{i_j}$
 - 6: $c_\zeta \leftarrow \frac{\alpha!}{\prod_{j=1}^n m_j} \prod_{j=1}^n \zeta_j$
 - 7: $\mathcal{S} \leftarrow \mathcal{S} \cup \{(c_\zeta, v_\zeta)\}$
 - 8: **end for**
 - 9: result $\leftarrow 0$
 - 10: **for all** $(c, v) \in \mathcal{S}$ **do**
 - 11: form the input jet $\mathbf{J}_0 \leftarrow (x, v, 0, \dots, 0)$ of length $p + 1$
 - 12: $\mathbf{J}_L \leftarrow$ jet propagation of $\mathbf{J}_0$ through every primitive of u using (14)–(16)
 - 13: $T_p(x; v) \leftarrow [\mathbf{J}_L]_p \quad \triangleright$ order- p coefficient of the output jet
 - 14: result $\leftarrow$ result $+ c \cdot T_p(x; v)$
 - 15: **end for**
 - 16: **return** result
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Remark 4.1. By Theorem 3.1 together with Theorem 3.6, Algorithm 1 returns $\partial^\alpha u(x)$ exactly (up to floating-point round-off in the jet propagation). The number of inner iterations is $|\mathcal{S}| = R_{\mathbb{C}}(z^\alpha)$, which is information-theoretically optimal over $\mathbb{C}$ (Theorem 3.5).

4.4 Cost and numerical considerations

For a fixed direction, the Taylor jet stores $p + 1$ coefficient tensors per activation and performs the same linear layers at each jet order; there is no chain of nested reverse-mode differentiation through the input coordinates. The total value-evaluation cost of B3 is therefore proportional to $R_{\mathbb{C}}(z^\alpha)$ directional jet evaluations, while B2 scales with the coalesced polarization count R_{pol} . The rank reduction is an algebraic cost reduction; the realized wall-clock time also depends on complex arithmetic, batching, activation kernels, and whether parameter gradients are required.

The schedule coefficients are roots of unity multiplied by $\alpha!/R_{\mathbb{C}}$, so coefficient magnitudes are explicit and do not come from solving an ill-conditioned linear system. The method has no finite-difference step size and no Monte Carlo variance. Remaining errors are ordinary floating-point round-off from Taylor recurrences, complex arithmetic, and the final summation over directions.

5 Numerical Experiments

We evaluate three claims:

- (C1) The Waring + Taylor-jet backend matches direct nested AD to round-off accuracy on the value of $\partial^\alpha u(x)$, and supports parameter-gradient back-propagation in PINN training.
- (C2) For repeated-index patterns where $R_{\mathbb{C}}$ is strictly smaller than the coalesced polarization count, the complex-Waring backend uses fewer Taylor directions; measured wall-clock differences are reported rather than inferred solely from rank.
- (C3) On a 4D manufactured sixth-order PDE, training a complex-parameter sinh PINN with the Waring backend has a much smaller time-per-step than nested AD and reaches the lowest error in the single-seed fixed-budget run reported here.

5.1 Experimental setup

Hardware. Single-GPU NVIDIA T4 (16 GB), CUDA 12.1, PyTorch 2.5.

Networks. The direct-derivative microbenchmark uses a real Linear-sinh MLP with hidden width 64 and depth 4, matching the saved run in `results/quick_compare_T4_20260602_0303.txt`. The PDE experiment uses a complex-parameter Linear-sinh MLP with hidden width 32 and depth 4. In the PDE loss, the physical solution is $\Re u$; imaginary parts of the weights are regularized via a Tikhonov term $\lambda \|\text{Im } W\|^2$ with $\lambda = 10^{-6}$.

Precision and reference. The microbenchmark runs in fp64, while the PDE training uses complex128 parameters and activations. Reference values for derivative accuracy are computed with backend B1 (direct nested AD) on the same network instance.

Timing protocol. The microbenchmark reports median wall-clock per derivative call over 60 timed repetitions after 5 warm-up calls, as recorded in the saved result file. The PDE experiment gives each backend a 60-second wall-clock training budget and reports the realized per-step time and peak allocated GPU memory. The saved PDE run uses one random seed shared across backends; it is intended as a controlled fixed-budget comparison, not as an uncertainty-quantified training study.

5.2 Direct derivative microbenchmark

We sample $B = 8$ inputs in $\mathbb{R}^d$ for $d = 8$ and measure the cost of computing one $\partial^\alpha u$ for each pattern in Table 1. Table 2 reports the wall-clock per call for the three backends on selected patterns.

pattern	p	R_C	R_{pol}	B1 (ms)	B2 (ms)	B3 (ms)
(3)	3	1	2	3.399	2.399	2.532
(4)	4	1	2	7.679	3.302	3.425
(2, 2)	4	3	4	7.689	3.314	3.458
(6)	6	1	3	52.253	5.916	5.837
(4, 2)	6	5	7	52.373	5.918	5.904
(2, 2, 2)	6	9	13	52.219	5.911	5.916
(1 ⁶)	6	32	32	52.323	5.961	5.943

Table 2: Median wall-clock per single $\partial^\alpha u$ call on the T4, with $d = 8$, $B = 8$, hidden width 64, depth 4, fp64, and a real-parameter sinh MLP. Times are from 60 timed repetitions after 5 warm-up calls. Bold entries mark patterns where $R_C < R_{\text{pol}}$. The largest reported relative error of a jet backend against B1 is below $6 \cdot 10^{-15}$.

The relative error of B2 and B3 against B1 is at the round-off level. At $p = 6$ the jet path is roughly $8.8\times$ faster than nested AD in this value-only benchmark. This verifies the accuracy part of C1 for the patterns tested; parameter-gradient support is exercised in the PINN experiment below.

5.3 Repeated-index regime

For repeated patterns, the algebraic advantage of B3 is a smaller number of Taylor directions: for example, (4, 2) uses 5 directions instead of 7, and (2, 2, 2) uses 9 instead of 13. Table 2 shows that this direction-count reduction does not automatically translate into a large value-only wall-clock gain on the tested T4 configuration: B2 and B3 are within a few percent, with the complex arithmetic overhead partly offsetting the shorter schedule. We therefore treat C2 as a complexity and direction-count advantage, and report measured timings case by case rather than claiming a universal wall-clock ordering.

5.4 PINN experiment: 4D manufactured sixth-order PDE

Problem. Consider the manufactured PDE on $\Omega = (-1, 1)^4$:

$$\partial_{x_1}^4 \partial_{x_2}^2 u(x) = f(x), \quad u|_{\partial\Omega} \text{ given}, \quad (19)$$

with $u_{\text{exact}}(x) = \sinh(x_1) \cos(x_2) e^{-(x_3^2 + x_4^2)/4}$. Since $\sinh^{(4)} = \sinh$ and $\cos^{(2)} = -\cos$, the source has the closed form $f = -u_{\text{exact}}$. The dominant operator is the single multi-index with active-exponent pattern (4, 2); by Theorem 3.5, $R_C = 5$ versus polarization count 7.

Architecture and training. We use the complex network specified in the experimental setup (hidden width 32, depth 4) with $d = 4$, input embedded as $x \mapsto x + 0i$. The PINN loss combines an interior residual term, a boundary-fit term, and a Tikhonov regularizer on the imaginary part of the weights:

$$\mathcal{L} = \mathcal{L}_{\text{int}} + \beta \mathcal{L}_{\text{bc}} + \lambda \sum_W \|\Im W\|^2, \quad (20)$$

with

$$\mathcal{L}_{\text{int}} = \mathbb{E}_{x \sim U(\Omega)} |\Re \partial^{(4,2)} u(x) - f(x)|^2, \quad \mathcal{L}_{\text{bc}} = \mathbb{E}_{x \sim U(\partial\Omega)} |\Re u(x) - u_{\text{exact}}(x)|^2,$$

$\beta = 10^2$ and $\lambda = 10^{-6}$. Training uses Adam with learning rate 10^{-3} ; each step samples $N_{\text{int}} = 128$ interior points and $N_{\text{bc}} = 64$ boundary points.

Comparison protocol. Each backend is given the same wall-clock budget of 60 seconds. We report *steps completed within the budget*, the wall-clock per step, peak GPU memory, the final value of $\mathcal{L}_{\text{int}}$ (averaged over the last 20 steps), and the relative L^2 error of $\Re u$ against u_{exact} on a held-out grid of size 4096. Results in Table 3 are measured directly on the T4 with the implementation of this paper. For a submission-quality accuracy comparison between B2 and B3, the same protocol should be repeated over multiple seeds with a fixed and recorded held-out evaluation set.

backend	steps	ms/step	peak (MB)	final $\mathcal{L}_{\text{int}}$	rel. L^2 error
B3 (complex Waring + jet)	4248	14.1	40	1.74×10^{-3}	8.50×10^{-3}
B2 (polarization + jet)	4192	14.3	49	6.32×10^{-3}	4.43×10^{-2}
B1 (direct nested AD)	211	285.6	103	2.82×10^{-1}	2.49×10^{-1}

Table 3: PINN training cost and accuracy on the 4D manufactured sixth-order problem (19), identical 60-second wall-clock budget per backend. Same network, optimizer, sampling rule and seed are used across rows. This is a single-seed run: the complex Waring backend reaches the lowest interior loss and the smallest L^2 error here, while nested AD completes only 211 steps and remains far from convergence.

Three observations are in order. First, the two jet-based backends share essentially identical per-step cost (14.1 and 14.3 ms), which is roughly $20\times$ faster than nested AD; nested AD’s peak memory is $2.5\times$ higher because of the deeper autograd graph. Second, B3 uses $R_{\mathbb{C}} = 5$ directions, two fewer than polarization ($R_{\text{pol}} = 7$). In this fixed-budget run it also reaches a smaller final L^2 error (8.5×10^{-3} vs. 4.4×10^{-2}), although this accuracy comparison is affected by stochastic sampling during training and should be read together with additional seeds in future work. Third, $\sinh$ is entire on $\mathbb{C}$, so complex Waring directions and complex-parameter $\sinh$ MLPs compose without any runtime dtype conversion in the inner loop. These results support C3 for this single-seed manufactured problem.

6 Benchmark problems

We evaluate the complex-parameter hyperbolic-sine network (“complex $\sinh$ ”), paired with the complex-Waring Taylor-jet residual backend of Section 4, on a suite of oscillatory and high-order partial differential equations. Every problem is a benchmark drawn from, or directly comparable to, the physics-informed neural network (PINN) literature; the exact manufactured solutions and parameters used in our runs are stated in full below, together with the source of each benchmark.

The suite is organized into three groups. The first probes *high-frequency second-order* problems, where spectral bias is the dominant difficulty (Sections 6.2–6.4). The second probes *high-order real* operators, where the derivative order is the dominant difficulty (Sections 6.5–6.8). The third probes *genuinely complex-valued* fields, where the unknown is naturally $\mathbb{C}$ -valued (Sections 6.9–6.10). Table 4 summarizes the suite.

6.1 Shared protocol

All architectures and training choices are held fixed across the suite so that the comparison isolates the effect of the network’s value field ($\mathbb{C}$ versus $\mathbb{R}$) rather than incidental tuning.

Table 4: Benchmark suite at a glance. Coordinates of a space–time problem are written (x, t) ; Δ is the spatial Laplacian; a, k, m are the swept parameters (wavenumber, mode, or temporal frequency). “Source” lists the benchmark from which the setting is taken or to which it is matched.

family	operator	order	exact solution	source
Helmholtz (high k)	$\Delta u + \kappa^2 u$	2	$\sin(a\pi x) \sin(a\pi y)$	[21]
Helmholtz (variable κ)	$\Delta u + \kappa^2(x)u$	2	$\sin(a\pi x) \sin(a\pi y)$	[5]
radial chirp	$-\Delta u + u$	2	$\sin(\frac{a\pi}{2}r^2)$	[17, 12]
polyharmonic	$\Delta^m u$	$2m$	$\sin(\pi x) [\sin(\pi y)]$	[20]
plate / beam	$\Delta^2 w$	4	$\sin(m\pi x) \sin(n\pi y)$	[20]
linearized KdV	$u_t + \delta u_{xxx}$	3	$\sin(k\pi x) \cos(k\pi t)$	[14]
Cahn–Hilliard	$u_t - M\Delta(u^3 - u - \gamma\Delta u [+ \kappa\Delta^2 u])$	4/6	$\sin(a\pi x) \cos(a\pi t)$	[14, 5]
cubic NLS	$iu_t + \frac{1}{2}u_{xx} + u ^2 u$	2	$\text{sech}(x) e^{ikt}$	[14]
Maxwell (lossy)	$\Delta E + \kappa^2 E, \kappa^2 \in \mathbb{C}$	2	$e^{ia\pi(x+y)}$	[9]

Architectures. The complex-sinh network is a multilayer perceptron with `complex128` parameters and the entire activation $\sinh$. It is compared against four real “high-frequency” baselines that are standard remedies for spectral bias: a random Fourier-feature network [17], a sinusoidal representation network (SIREN) [16], a multi-scale deep network (MscaleDNN) [12], and plain real $\tanh$ and $\sinh$ baselines. For genuinely complex-valued targets (Sections 6.9–6.10) each real baseline carries the field as a split real–imaginary pair (an $\mathbb{R}$ -valued PINN, RVPINN), as in the original complex-valued PINN formulation [14].

Parameter matching. A width- H complex network has $\approx 2H^2$ real degrees of freedom per layer. To make the contest an equal-parameter one, every single real baseline is widened to $H_{\mathbb{R}} = \lceil \sqrt{2}H \rceil$ (so it has at least the real parameter count of the complex network); for the split-real pairs each component uses width H so that the *pair* matches the complex network.

Frequency matching. Frequency-aware architectures are initialized at the problem’s characteristic wavenumber, so each method receives its best starting point at every swept frequency: the SIREN / complex-sinh first-layer scale ω_0 and the Fourier-feature bandwidth σ are set proportional to the target spatial frequency of each case (specified per family below). For the high-order families the initialization scale is held *at* the target frequency rather than *above* it, because an order- m operator amplifies an initialization frequency ω like ω^m ; an over-large ω_0 is amplified by ω_0^m and buries the learning signal.

Derivative backend. All jet-compatible architectures (complex $\sinh$, real $\sinh/\tanh$, SIREN, Fourier, MscaleDNN) evaluate every residual derivative with the *same* complex-Waring Taylor-jet operator of Section 4, so the derivative machinery does not bias the comparison. Nonlinear fluxes are written purely from single-monomial partials of the network: for example the Cahn–Hilliard term

$$\Delta(u^3) = 3u^2 u_{xx} + 6u (u_x)^2 \quad (21)$$

uses only u, u_x, u_{xx} , each a single Taylor-jet evaluation.

Optimization and metric. Training uses Adam with a wall-clock budget shared across all methods, a time-based cosine learning-rate schedule decaying from 10^{-3} to a floor of 10^{-4} , and a fixed dense collocation set. The reported accuracy is the relative L^2 error of $\text{Re } u$ (or of u for

complex-valued targets) against the exact solution on a held-out grid of 8192 points,

$$\mathcal{E}_{L^2} = \frac{\|u_\theta - u_\star\|_{L^2(\Omega)}}{\|u_\star\|_{L^2(\Omega)}}, \quad (22)$$

averaged over independent random seeds. Boundary and initial data are imposed by a penalty term with weight 100; for the polyharmonic and plate families the simply-supported (Navier) conditions are enforced on the lower even derivatives (see Sections 6.5–6.6).

6.2 High-wavenumber Helmholtz

The first benchmark is the constant-coefficient Helmholtz equation on the unit square,

$$\Delta u + \kappa^2 u = f \quad \text{in } \Omega = (-1, 1)^2, \quad \kappa = a\pi, \quad (23)$$

with homogeneous Dirichlet data $u = 0$ on $\partial\Omega$. The exact solution is the separable eigenmode

$$u_\star(x, y) = \sin(a\pi x) \sin(a\pi y), \quad (24)$$

and the source follows in closed form,

$$\begin{aligned} f &= \Delta u_\star + \kappa^2 u_\star \\ &= (-2(a\pi)^2 + (a\pi)^2) u_\star = -(a\pi)^2 u_\star. \end{aligned} \quad (25)$$

The wavenumber is swept over $a \in \{2, 4, 6, 8, 10\}$, so the number of half-periods across the domain grows while the operator order stays fixed at two; this isolates the effect of frequency on each architecture.

Provenance and comparability. This is the Helmholtz benchmark of Wang, Teng and Perdikaris [21], who use $\Delta u + k^2 u = q$ on $(-1, 1)^2$ with $u = \sin(a_1 \pi x) \sin(a_2 \pi y)$ and the matching source $q = (k^2 - (a_1 \pi)^2 - (a_2 \pi)^2) u$; the same family is the Poisson–Boltzmann/Helmholtz task in the PINNacle suite [5]. We take the isotropic case $a_1 = a_2 = a$ with $k = a\pi$, and additionally report the anisotropic case $(a_1, a_2) = (1, 4)$ of [21] verbatim (with $k = 1$, $q = (k^2 - (a_1 \pi)^2 - (a_2 \pi)^2) u_\star$) as a one-to-one numerical comparison.

Setting. $\Omega = (-1, 1)^2$; Dirichlet $u = 0$; isotropic sweep $a \in \{2, 4, 6, 8, 10\}$ plus the anisotropic $(1, 4)$ case; $\omega_0 = \max(10, 2\pi a)$, $\sigma = \max(2, \pi a)$; residual scale $(a\pi)^2$.

6.3 Variable-coefficient (scattering) Helmholtz

To move from a single clean eigenmode to a heterogeneous medium, we let the zeroth-order coefficient vary in space,

$$\Delta u + \kappa^2(x) u = f \quad \text{in } \Omega = (-1, 1)^2, \quad (26)$$

with a smooth lens-like contrast

$$\kappa^2(x, y) = (a\pi)^2 (1 + C \sin(\pi x) \sin(\pi y)), \quad C = 0.5. \quad (27)$$

The manufactured solution and Dirichlet data are again (24) with $u = 0$ on $\partial\Omega$, and the source is

$$f = -2(a\pi)^2 u_\star + \kappa^2(x) u_\star. \quad (28)$$

The background wavenumber is swept over $a \in \{2, 4, 6\}$.

Provenance and comparability. Variable-coefficient (scattering) Helmholtz problems in a heterogeneous medium are the realistic electromagnetics/acoustics regime represented in the PINNacle Poisson–Boltzmann/Helmholtz tasks [5]; the $\pm 50\%$ smooth contrast here is a controlled, exactly-known-solution instance of that regime. It also exercises the variable-coefficient path of the residual assembly, in which $\kappa^2(x)$ is precomputed once on the collocation set.

Setting. $\Omega = (-1, 1)^2$; Dirichlet $u = 0$; contrast $C = 0.5$; sweep $a \in \{2, 4, 6\}$; ω_0, σ as in Section 6.2.

6.4 Non-separable radial chirp

The eigenmode benchmarks above are separable products of sines, which sine-based real baselines can represent almost natively and which may therefore mask any expressivity gap. To remove this confound we use a non-separable target with a space-varying local frequency,

$$-\Delta u + u = f \quad \text{in } \Omega = (-1, 1)^2, \quad (29)$$

$$u_\star(x, y) = \sin(\phi(x, y)), \quad \phi(x, y) = \frac{a\pi}{2} (x^2 + y^2), \quad (30)$$

a radial chirp whose instantaneous frequency grows with radius,

$$|\nabla \phi| = a\pi r, \quad r = \sqrt{x^2 + y^2}. \quad (31)$$

The source is assembled from

$$\begin{aligned} \Delta u_\star &= -(a\pi)^2 r^2 \sin \phi + 2a\pi \cos \phi, \\ f &= -\Delta u_\star + u_\star, \end{aligned} \quad (32)$$

with Dirichlet data $u = u_\star$ on $\partial\Omega$ and the sweep $a \in \{2, 4, 6, 8\}$.

Provenance and comparability. Space-varying-frequency (chirp) targets are the standard stress test for high-frequency representation methods such as Fourier features [17] and multi-scale networks [12], which are tuned for stationary spectra. This benchmark is the fairness control of the suite: every architecture must *construct* the varying oscillation rather than read it off a fixed frequency basis.

Setting. $\Omega = (-1, 1)^2$; Dirichlet $u = u_\star$; sweep $a \in \{2, 4, 6, 8\}$; ω_0, σ as in Section 6.2.

6.5 Polyharmonic order sweep

To isolate the effect of the differential *order* from that of frequency, we fix the solution and frequency and vary only the order of a polyharmonic operator. The eigenmode identity $\Delta^m u = (-S)^m u$ holds for the sinusoidal eigenfunctions below, giving

$$\Delta^m u = (-S)^m u, \quad \begin{cases} u_\star = \sin(\pi x) \sin(\pi y), & S = 2\pi^2 \text{ (2D)}, \\ u_\star = \sin(\pi x), & S = \pi^2 \text{ (1D)}. \end{cases} \quad (33)$$

Simply-supported (Navier) boundary conditions are imposed exactly through the lower even derivatives,

$$\Delta^j u = 0 \quad \text{on } \partial\Omega, \quad j = 0, 1, \dots, m-1. \quad (34)$$

The order is swept over $2m \in \{2, 4, 6\}$ in two dimensions and $2m \in \{2, 4, 6, 8, 10\}$ in one dimension. In one dimension the operator is a single monomial partial ∂_x^{2m} , which is cheap enough to reach order ten; in two dimensions Δ^m expands into $m+1$ high-order terms.

Provenance and comparability. Biharmonic and polyharmonic operators are the canonical high-order PINN testbed [20]; the biharmonic ($m = 2$) and triharmonic ($m = 3$) cases coincide with the polyharmonic family used in randomized-smoothing PINNs. Holding u_* , the frequency and the domain fixed while varying only m is our isolation design for the order axis.

Setting. $\Omega = (-1, 1)^d$, $d \in \{1, 2\}$; Navier BC (34); 1D $\omega_0 = \sigma = \pi$, 2D $\omega_0 = 2\pi$, $\sigma = \pi$; residual scale S^m .

6.6 Kirchhoff plate and Euler–Bernoulli beam

The fourth-order vibration eigenproblems of structural mechanics give two oscillatory biharmonic benchmarks. For the Kirchhoff plate,

$$\Delta^2 w = S^2 w \quad \text{in } \Omega = (-1, 1)^2, \quad S = (m^2 + n^2)\pi^2, \quad (35)$$

with simply-supported edges $w = 0$ and $\Delta w = 0$ on $\partial\Omega$ and the mode shape

$$w_*(x, y) = \sin(m\pi x) \sin(n\pi y). \quad (36)$$

For the Euler–Bernoulli beam in one dimension,

$$w'''' = (m\pi)^4 w \quad \text{in } (-1, 1), \quad w_* = \sin(m\pi x), \quad (37)$$

with simply-supported ends $w = 0$ and $w'' = 0$. We sweep the mode number, using the isotropic plate modes (m, m) with $m \in \{1, 2, 3\}$, the anisotropic mixed modes $(m, m + 1)$ with $m \in \{2, 3, 4\}$, and the beam modes $m \in \{1, 2, 3\}$. The order is fixed at four; the sweep raises the oscillation only.

Provenance and comparability. The biharmonic plate and beam are standard PINN structural benchmarks: Vahab et al. [20] solve the simply-supported Kirchhoff plate ($D \Delta^2 w = q$), and the simply-supported Euler–Bernoulli beam with mode shape $\sin(m\pi x)$ is the textbook 1D specialization. We use the free-vibration eigenmode form $\Delta^2 w = S^2 w$ with the corresponding manufactured right-hand side, so the exact solution is known in closed form.

Setting. Plate $\Omega = (-1, 1)^2$, Navier BC $w = \Delta w = 0$; beam $(-1, 1)$, Navier BC $w = w'' = 0$; modes (m, m) , $(m, m+1)$, and beam m ; $\omega_0 = \max(10, 2\pi f)$, $\sigma = \max(2, \pi f)$ with $f = \max(m, n)$; residual scale S^2 .

6.7 Linearized Korteweg–de Vries / dispersive wave

The third-order dispersive operator is exercised by a linearized KdV (Airy) equation in space–time,

$$u_t + \delta u_{xxx} = f \quad \text{in } (x, t) \in (-1, 1)^2, \quad \delta = 1, \quad (38)$$

with the oscillatory travelling profile

$$u_*(x, t) = \sin(k\pi x) \cos(k\pi t) \quad (39)$$

and Dirichlet data $u = u_*$ on the space–time boundary. The source is

$$\begin{aligned} f &= u_{*,t} + \delta u_{*,xxx} \\ &= -k\pi \sin(k\pi x) \sin(k\pi t) - (k\pi)^3 \cos(k\pi x) \cos(k\pi t), \end{aligned} \quad (40)$$

and the wavenumber is swept over $k \in \{2, 3, 4, 5, 6\}$.

Provenance and comparability. KdV is a canonical PINN benchmark; Raissi et al. [14] use the nonlinear form $u_t + \lambda_1 u u_x + \lambda_2 u_{xxx} = 0$ with $\lambda_2 = 0.0025$. We use the linear dispersive (Airy) operator with a manufactured oscillatory solution in order to isolate the odd third-order term u_{xxx} from the convective nonlinearity; the canonical nonlinear KdV can be added as a separate case for a direct comparison.

Setting. $(x, t) \in (-1, 1)^2$; Dirichlet $u = u_*$; $\delta = 1$; sweep $k \in \{2, 3, 4, 5, 6\}$; $\omega_0 = \max(10, 2\pi k)$, $\sigma = \max(2, \pi k)$.

6.8 Cahn–Hilliard

The phase-field Cahn–Hilliard equation combines a high-order linear operator with a cubic nonlinearity. In one space dimension plus time,

$$u_t = M[\Delta(u^3) - \Delta u - \gamma \Delta^2 u] \quad (4\text{th order}), \quad (41)$$

$$u_t = M[\Delta(u^3) - \Delta u - \gamma \Delta^2 u + \kappa \Delta^3 u] \quad (6\text{th order}), \quad (42)$$

with $M = \gamma = \kappa = 1$, $\Delta = \partial_x^2$, and $(x, t) \in (-1, 1)^2$. The manufactured solution is

$$u_*(x, t) = \sin(a\pi x) \cos(a\pi t), \quad (43)$$

the nonlinear flux is expanded as in (21), and the source term and the boundary targets (on u , u_{xx} , and additionally u_{xxxx} for the sixth-order case) are obtained from the analytic solution. We use $a \in \{2, 3\}$ at orders 4 and 6.

Provenance and comparability. Cahn–Hilliard is a standard fourth-order phase-field PINN benchmark, and a sixth-order variant is used in randomized-smoothing PINNs; chaotic/high-order space–time PDEs of this type (Kuramoto–Sivashinsky, Gray–Scott) form the nonlinear high-order tier of PINNacle [5]. The manufactured solution (43) gives an exactly known reference while retaining the cubic nonlinearity and the genuine fourth/sixth-order operator.

Setting. $(x, t) \in (-1, 1)^2$; $M = \gamma = \kappa = 1$; orders $\{4, 6\}$; sweep $a \in \{2, 3\}$; $\omega_0 = \max(10, 2\pi a)$, $\sigma = \max(2, \pi a)$.

6.9 Cubic nonlinear Schrödinger

The first genuinely complex-valued benchmark is the cubic nonlinear Schrödinger (NLS) equation,

$$i u_t + \frac{1}{2} u_{xx} + |u|^2 u = f, \quad u : \Omega \rightarrow \mathbb{C}. \quad (44)$$

We use the standing bright-soliton family as the manufactured solution,

$$u_*(x, t) = \text{sech}(x) e^{ikt}, \quad (45)$$

for which a direct computation gives the source

$$i u_{*,t} + \frac{1}{2} u_{*,xx} + |u_*|^2 u_* = \left(-k + \frac{1}{2} - \text{sech}^2 x + \text{sech}^2 x \right) u_* = \left(\frac{1}{2} - k \right) u_*, \quad (46)$$

so $f = (\frac{1}{2} - k) u_*$, and at $k = \frac{1}{2}$ the source vanishes and u_* is the exact bright soliton. The temporal frequency is swept over $k \in \{1, 2, 4\}$. A single complex-sinh network carries u in $\mathbb{C}$ natively; the real baselines carry u as a split real/imaginary pair, parameter-matched as a pair.

Provenance and comparability. This is the complex-valued benchmark introduced by Raissi et al. [14], whose canonical setting is

$$i h_t + \frac{1}{2} h_{xx} + |h|^2 h = 0, \quad x \in [-5, 5], \quad t \in [0, \pi/2], \quad h(0, x) = 2 \operatorname{sech}(x), \quad (47)$$

with periodic boundary conditions. We use the same domain window $x \in [-5, 5]$, $t \in [0, \pi/2]$, with the manufactured standing soliton (45) in place of the free amplitude-2 evolution, so the reference solution is known in closed form while the equation, the cubic nonlinearity, and the complex-valued field are identical to the canonical setting.

Setting. $x \in [-5, 5]$, $t \in [0, \pi/2]$ (inputs normalized to $[-1, 1]^2$; chain-rule factors $1/L_x, 1/L_t$); Dirichlet/initial penalty; sweep $k \in \{1, 2, 4\}$; complex vs. split-real representation.

6.10 Time-harmonic Maxwell (lossy medium)

The second complex-valued benchmark is the time-harmonic transverse-magnetic (TM_z) Maxwell system, which for the out-of-plane field reduces to a complex Helmholtz equation,

$$\Delta E + \kappa^2 E = f \quad \text{in } \Omega = (-1, 1)^2, \quad \kappa^2 = (a\pi)^2 (1 + i\beta), \quad (48)$$

with loss tangent $\beta = 0.2$. The complex permittivity makes the field genuinely complex-valued, and the manufactured solution is a traveling plane wave,

$$E_\star(x, y) = e^{ia\pi(x+y)}, \quad (49)$$

with Dirichlet data $E = E_\star$ on $\partial\Omega$ and source

$$f = (-2(a\pi)^2 + \kappa^2) E_\star. \quad (50)$$

The wavenumber is swept over $a \in \{2, 4, 6\}$. As in the NLS case, the complex network carries E natively while the real baselines use a split-real pair.

Provenance and comparability. Time-harmonic Maxwell / frequency-domain electromagnetic scattering is an inherently complex-valued PDE family for which complex-valued surrogates have shown clear advantages over real-valued ones [9]. The lossy ($i\beta$) term is the linear complex-valued companion to the cubic NLS test: it forces a genuinely complex field through a linear second-order operator.

Setting. $\Omega = (-1, 1)^2$; Dirichlet $E = E_\star$; loss tangent $\beta = 0.2$; sweep $a \in \{2, 4, 6\}$; $\omega_0 = \max(10, 2\pi a)$, $\sigma = \max(2, \pi a)$; complex vs. split-real representation.

7 Complex versus real networks on the oscillatory suite

Section 6 fixed the problems and the protocol. We now report the outcome of the contest between the complex-parameter sinh network and the real high-frequency baselines. The question is deliberately narrow: *at a matched real-parameter budget and a shared training protocol, does carrying the network’s value field in $\mathbb{C}$ buy accuracy on oscillatory and high-order PDEs?*

Reading the tables. Every entry is the relative L^2 error (22) on the held-out grid, averaged over the random seeds; the best (smallest) entry in each row is set in bold. For each problem family we also report the *advantage factor*

$$\text{adv} = \frac{\min_{\text{real baselines}} \mathcal{E}_{L^2}}{\mathcal{E}_{L^2}(\text{complex sinh})}, \quad (51)$$

so that $\text{adv} > 1$ means the complex network beats the best real contender on that instance, and the win count over the family quantifies how often this happens.

Hardware and budget. All runs in this section are measured on a single NVIDIA H20 GPU (97 GB, CUDA 12.1, PyTorch 2.5) in `float64/complex128`. Each (problem, variant, seed) receives the same wall-clock budget (60–85 s depending on family) and the cosine schedule of Section 6.1; reported errors are means over two to three seeds. The complete raw logs and per-seed CSV files are in `results/`.

7.1 High-frequency second-order problems

Here the obstacle is spectral bias rather than derivative order. Table 5 collects the constant-coefficient Helmholtz frequency sweep, its variable-coefficient “scattering” variant, and the non-separable radial chirp. The complex sinh network is the most accurate method on every constant-coefficient Helmholtz wavenumber and on every chirp frequency; on the heterogeneous (variable- κ) problem it leads only at the highest wavenumber, with a Fourier-feature or SIREN network ahead at low frequency. The single anisotropic case $(a_1, a_2) = (1, 4)$ is also lost: at this low, strongly directional frequency a SIREN reaches 2.2×10^{-1} against the complex network’s 3.9×10^{-1} , consistent with SIREN’s strength on separable low-frequency sinusoids. The chirp is the cleanest expressivity signal: because its solution is not a product of one-dimensional sines, the sine-based MscaleDNN – competitive on separable targets – degrades sharply (errors above one), while the complex network retains the smallest error throughout.

Table 5: High-frequency second-order problems: relative L^2 error (22) (lower is better; best in bold). a is the swept wavenumber.

family	a	complex sinh	Fourier	SIREN	MscaleDNN
Helmholtz	2	5.22e-2	1.52e-1	8.09e-2	6.99e-1
	4	3.22e-1	4.32e-1	3.46e-1	1.00e0
	6	5.71e-1	8.51e-1	8.19e-1	1.00e0
	8	6.29e-1	9.32e-1	9.27e-1	1.00e0
	10	7.71e-1	9.86e-1	9.83e-1	1.00e0
Helmholtz (anis. 1, 4)	–	3.89e-1	7.79e-1	2.21e-1	1.00e0
Helmholtz (var. κ)	2	3.59e-1	2.11e-1	3.15e-1	1.00e0
	4	4.64e-1	5.45e-1	4.37e-1	1.00e0
	6	7.12e-1	8.72e-1	8.07e-1	1.00e0
radial chirp	2	3.76e-1	7.07e-1	8.74e-1	7.93e-1
	4	4.52e-1	9.19e-1	7.16e-1	1.60e0
	6	8.47e-1	1.18e0	9.76e-1	1.89e0
	8	8.30e-1	1.24e0	9.64e-1	2.39e0

7.2 High-order real operators

Table 6 reports the order-controlled families. The one-dimensional polyharmonic sweep isolates the differentiation order on a single monomial ∂_x^{2m} : the complex network is one to two orders of magnitude more accurate than every real baseline at orders two and four, retains the lead at order eight, and is edged out by MscaleDNN only at order six; at order ten the fixed budget is insufficient for any method. In two dimensions the gap widens with order, reaching a $13\times$ advantage on the order-six triharmonic where all real baselines have effectively failed. The anisotropic plate (Δ^2 , mixed modes $(m, m+1)$) is won at every mode by a clear margin, and the isotropic plate is won at the higher modes with an advantage that grows from parity to $1.85\times$. The one-dimensional Euler–Bernoulli beam is the subtler case: a Fourier-feature network ties or wins on the smoothest modes, because a one-dimensional fourth-order sine is precisely the stationary single-frequency signal that Fourier features are designed for. The exception

is the third-order linearized KdV: there a SIREN network is the most accurate at the higher wavenumbers, and the complex network leads only at one wavenumber, reflecting that a first-order-in-time dispersive operator does not present the balanced repeated-index structure that the complex ansatz exploits. The nonlinear Cahn–Hilliard phase field returns to the favourable regime: the complex network wins three of the four instances, including both sixth-order cases, and trails a Fourier network only on the smoothest fourth-order amplitude.

Table 6: High-order real operators: relative L^2 error (22). “order” is the operator order; the swept parameter is the order $2m$ (polyharmonic), the plate mode m , or the KdV wavenumber k . The plain real sinh baseline (omitted) exceeds 0.85 on every high-order row.

family	sweep	complex sinh	Fourier	SIREN	MscaleDNN
polyharmonic 1D	$2m=2$	4.15e-5	2.33e-4	2.13e-4	3.09e-4
	$2m=4$	2.03e-3	1.20e-2	3.15e-2	6.83e-2
	$2m=6$	1.42e-1	6.90e-1	9.66e-1	1.25e-1
	$2m=8$	2.14e-1	1.00e0	1.00e0	4.08e-1
	$2m=10$	9.95e-1	1.00e0	1.01e0	9.98e-1
polyharmonic 2D	$2m=2$	4.23e-4	6.66e-4	7.83e-4	1.09e-2
	$2m=4$	3.96e-2	3.17e-2	9.83e-1	8.99e-1
	$2m=6$	7.67e-2	9.99e-1	9.94e-1	9.96e-1
plate (iso)	$m=1$	4.62e-2	4.24e-2	9.88e-1	9.80e-1
	$m=2$	2.47e-1	2.76e-1	5.35e-1	1.00e0
	$m=3$	3.63e-1	6.72e-1	9.24e-1	1.00e0
plate (mixed)	$m=2$	6.59e-1	9.78e-1	9.39e-1	1.00e0
	$m=3$	5.54e-1	9.93e-1	9.81e-1	1.00e0
	$m=4$	6.90e-1	9.97e-1	9.88e-1	1.00e0
beam (1D)	$m=1$	4.36e-2	1.35e-2	1.01e0	9.78e-2
	$m=2$	1.29e-1	1.57e-1	6.11e-1	5.49e-1
	$m=3$	2.22e-1	1.16e-1	8.06e-1	1.68e0
linearized KdV	$k=2$	3.65e-1	2.89e-1	4.54e-1	1.29e0
	$k=3$	5.68e-1	6.39e-1	6.05e-1	2.62e0
	$k=4$	8.70e-1	7.22e-1	6.55e-1	2.51e0
	$k=5$	9.48e-1	1.06e0	6.82e-1	4.03e0
	$k=6$	8.03e-1	1.10e0	7.37e-1	3.12e0
Cahn–Hilliard	$2m=4, a2$	4.45e-1	3.65e-1	5.55e-1	1.30e0
	$2m=4, a3$	4.10e-1	5.90e-1	7.68e-1	1.51e0
	$2m=6, a2$	5.60e-1	8.02e-1	9.74e-1	1.33e0
	$2m=6, a3$	8.34e-1	9.11e-1	9.99e-1	1.57e0

7.3 Genuinely complex-valued fields

For complex-valued targets the real baselines carry the field as a split real–imaginary pair (an $\mathbb{R}$ -valued PINN). Table 7 collects the cubic nonlinear Schrödinger soliton and the time-harmonic Maxwell problem in a lossy medium. The complex sinh network is the most accurate method on every instance of both families: it leads the cubic NLS by a factor of 1.6–2.2 over the best split-real network and reaches a relative error below 10^{-2} at $k = 1$, and on lossy Maxwell the split-real tanh pair fails outright (errors above 1.5). This is the regime where a complex value field is most natural, and the measured advantage is the most consistent in the whole suite.

Table 7: Genuinely complex-valued problems: relative L^2 error (22). Real baselines carry a split real–imaginary pair.

family	k / a	complex sinh	SIREN	Fourier	split tanh
cubic NLS	$k=1$	7.56e-3	1.20e-2	1.82e-2	2.76e-1
	$k=2$	3.26e-2	7.06e-2	7.90e-2	3.84e-1
	$k=4$	4.13e-2	8.82e-2	8.95e-2	3.95e-1
Maxwell (lossy)	$a=2$	2.31e-1	3.63e-1	3.94e-1	1.76e0
	$a=4$	4.84e-1	7.62e-1	8.29e-1	1.82e0
	$a=6$	6.22e-1	9.38e-1	1.02e0	1.55e0

7.4 Summary and acceptance criterion

Table 8 aggregates the contest by family. Counting an instance as a win when the complex network attains the strictly smallest relative L^2 error, the complex sinh ansatz is the best of the field on 32 of the 45 instances of this section. The wins are not uniform: they concentrate on the regimes the ansatz is designed for – balanced high-order operators (the polyharmonic, plate, and Cahn–Hilliard families) and genuinely complex-valued fields (NLS and Maxwell, where the complex network wins every instance) – and on non-separable high-frequency targets (the chirp), while the gains shrink or reverse on the first-order-in-time dispersive KdV, on the one-dimensional beam and low-frequency plate modes, and on the low-frequency or strongly directional Helmholtz problems.

We had set the stronger *acceptance criterion* that the advantage factor (51) grow monotonically with order or wavenumber. The data do not support this in its strict form: although the per-family advantage is almost always above one, it does not increase monotonically, and at the most extreme order or wavenumber every method saturates near $\mathcal{E}_{L^2} \approx 1$ within the fixed budget, so the ratio is no longer informative. The honest reading is therefore the weaker but robust one: *at matched parameters and a shared protocol, a complex value field is competitive-to-superior across the oscillatory suite and clearly superior on balanced high-order and complex-valued problems, but it is not a universal win and does not overcome the budget-limited saturation of high-frequency PINNs.*

Table 8: Per-family summary. “wins” counts instances on which the complex sinh network attains the smallest relative L^2 error; “adv” is the range of the advantage factor (51) over the family’s instances.

family	regime	wins	adv range
Helmholtz (high k)	high-freq. 2nd order	5/5	1.07–1.55
Helmholtz (anis. 1, 4)	directional 2nd order	0/1	0.57
Helmholtz (var. κ)	heterogeneous 2nd order	1/3	0.59–1.13
radial chirp	non-separable 2nd order	4/4	1.15–1.88
polyharmonic 1D	order sweep $2m \leq 10$	4/5	0.88–5.94
polyharmonic 2D	order sweep $2m \leq 6$	2/3	0.80–12.96
plate (isotropic)	4th order, modes	2/3	0.92–1.85
plate (mixed modes)	4th order, anisotropic	3/3	1.42–1.77
beam (1D)	4th order, 1D	1/3	0.31–1.22
linearized KdV	3rd order, dispersive	1/5	0.72–1.07
Cahn–Hilliard	4th/6th, nonlinear	3/4	0.82–1.44
cubic NLS	complex-valued	3/3	1.59–2.16
Maxwell (lossy)	complex-valued	3/3	1.51–1.57
total		32/45	

8 Discussion and Limitations

Strengths. Complex Waring is at its strongest when the active-exponent pattern is repeated and balanced, e.g., $(2, 2)$, $(2, 2, 2)$, $(3, 3)$, $(4, 2)$. For these the rank $\prod_{j \geq 1} (a_j + 1)$ is much smaller than 2^{p-1} ; concretely, $(2, 2, 2)$ yields rank 9 while polarization needs 13 directions for the same multi-index. PINNs whose residual contains such a single dominant operator – not uncommon in fourth- and sixth-order beam, plate, and Cahn–Hilliard models – benefit directly.

Limitations. Square-free patterns saturate the 2^{p-1} bound over both $\mathbb{R}$ and $\mathbb{C}$; for $\partial_x \partial_y \partial_z \partial_w \partial_u \partial_v$ the method offers no rank advantage over polarization. Operators that are *sums* of multi-indices (Laplacian, biharmonic) fall outside our scope and are best served by trace-collapsing methods [11, 3].

Experimental evidence. The direct-derivative benchmark is a value-mode accuracy and timing test; it verifies round-off agreement and exposes the direction-count advantage, but it does not by itself prove a universal wall-clock ordering between complex Waring and real polarization. The PINN experiment is a single-seed fixed-budget run. It demonstrates that the jet backends reduce per-step cost substantially relative to nested AD on the manufactured problem, while the observed accuracy gap between B3 and B2 should be validated with multiple seeds, fixed evaluation grids, and structured CSV/JSON result files before being used as a statistical comparison.

Complex value fields on oscillatory PDEs. The oscillatory study (Sections 6–7) gives a second, independent reason to value the entire-on- $\mathbb{C}$ backend: it makes complex-parameter sinh networks usable as a PINN ansatz, and that ansatz is empirically more expressive than strong real high-frequency baselines on most of a literature-grounded suite. The advantage is real but bounded. It is largest exactly where the theory predicts – balanced repeated-index high-order operators, where the complex schedule is also rank-optimal – and on genuinely complex-valued fields, where a split real–imaginary surrogate is at a structural disadvantage. It shrinks or disappears on first-order-in-time dispersive operators (KdV) and on low-frequency heterogeneous media, and no representation escapes the budget-limited saturation of high-frequency PINNs at the most extreme orders and wavenumbers. We therefore present complex sinh networks as a competitive default for high-order and complex-valued oscillatory PINNs, not as a universal replacement for real architectures.

Connection to STDE. The Stochastic Taylor Derivative Estimator [15] solves a strictly broader problem – arbitrary contraction of the order- k derivative tensor – via stochastic sparse jets driven by integer partitions. For one fixed multi-index α our deterministic Waring schedule is exact, while STDE is an unbiased estimator with variance dictated by its sample budget. The two are complementary: STDE is the natural choice when the operator is a linear combination of many multi-indices; Waring is the natural choice when attention is on one dominant multi-index that one wishes to evaluate exactly.

9 Conclusion

We have shown that the construction of a directional schedule for one mixed partial derivative $\partial^\alpha u$ of order p is exactly the problem of a Waring decomposition of the monomial z^α . The classical Carlini–Catalisano–Geramita rank theorem and its roots-of-unity construction yield an explicit, rank-optimal directional schedule over $\mathbb{C}$; paired with Taylor-mode propagation through sinh-activated MLPs, this gives a deterministic, exact, parameter-differentiable backend for high-order single-monomial residuals. In the tested 4D manufactured sixth-order PDE, the

jet backends are much faster per step than nested automatic differentiation, and the complex Waring backend achieves the lowest error in the reported single-seed fixed-budget run.

The method is at its most useful for repeated-index multi-indices, where the complex Waring rank is materially smaller than the coalesced real polarization count. For square-free patterns it offers no rank advantage; for operator sums the rank is not shortened, although the jet still evaluates them exactly term by term. Within its niche – a single, dominating, repeated-index high-order operator – it provides an exact deterministic alternative to stochastic estimators, while leaving broader operator contractions to methods designed for that setting.

The same entire-on- $\mathbb{C}$ backend has a second payoff. Because it makes complex-parameter sinh networks high-order differentiable at no extra inner-loop cost, it enables a complex value field as a PINN ansatz. On a literature-grounded suite of nine oscillatory and high-order PDE families, evaluated at matched real-parameter counts and under one shared training protocol against Fourier-feature, SIREN, and MscaleDNN baselines, the complex sinh network is the most accurate method on the majority of instances and is clearly strongest on balanced high-order operators and on genuinely complex-valued problems. It is not a universal win – dispersive and low-frequency heterogeneous problems remain better served by real high-frequency networks, and all methods saturate at the most extreme frequencies within a fixed budget – but it is a competitive default precisely in the high-order and complex-valued regimes that motivate this work.

A Proofs of Directional Schedule Identities

A.1 Roots-of-unity filter

Lemma A.1. *For any positive integer m and integer k , $\sum_{\zeta: \zeta^m=1} \zeta^k = m \mathbf{1}_{\{m|k\}}$.*

Proof. If $m \mid k$ every term is 1; otherwise the sum is a non-trivial geometric series in the primitive m -th root of unity, which sums to zero. $\square$

A.2 Proof of Theorem 3.6

We verify condition (2) of Theorem 3.1: namely that the schedule (9) satisfies $p! z^\alpha = \sum_{\zeta} c_{\zeta} (v_{\zeta} \cdot z)^p$ in $\mathbb{C}[z]_p$.

Expanding $(v_{\zeta} \cdot z)^p$ by the multinomial theorem and collecting the coefficient of z^{β} (with $|\beta| = p$):

$$[z^{\beta}] \sum_{\zeta} c_{\zeta} (v_{\zeta} \cdot z)^p = \frac{\alpha!}{\prod_{j=1}^n m_j} \binom{p}{\beta} \prod_{j=1}^n \left(\sum_{\zeta_j^{m_j}=1} \zeta_j^{1+\beta_j} \right).$$

By Lemma A.1, each inner sum equals m_j iff $m_j \mid 1 + \beta_j$, i.e. $\beta_j \equiv a_j \pmod{m_j}$, and equals zero otherwise.

A short counting argument shows that the unique feasible joint assignment is $\beta = \alpha$. Indeed, the congruence forces $\beta_j = a_j + t_j(a_j + 1)$ for $j \geq 1$ with $t_j \geq 0$. If some $t_j > 0$, then

$$\beta_0 = a_0 - \sum_{j=1}^n t_j(a_j + 1) < 0,$$

because $a_j \geq a_0$, impossible. Hence all $t_j = 0$ and $\beta = \alpha$. With $\beta = \alpha$ all the m_j factors cancel and the leading constant becomes $\alpha! \binom{p}{\alpha} = p!$, giving $[z^{\alpha}] \sum_{\zeta} c_{\zeta} (v_{\zeta} \cdot z)^p = p!$ and all other coefficients vanish. $\square$

A.3 Apolar pairing alternative

The Waring rank $R_{\mathbb{C}}(F)$ of a homogeneous polynomial $F \in \mathbb{C}[z]_p$ admits the apolar characterization $R_{\mathbb{C}}(F) = \min\{\deg X : X \subset \mathbb{P}^n \text{ a finite reduced scheme apolar to } F\}$, where X is apolar to F iff $I_X \subset F^\perp$ and $F^\perp$ is the apolar (Macaulay inverse) ideal of F . For $F = z^\alpha$, $F^\perp = (\partial_0^{a_0+1}, \dots, \partial_n^{a_n+1})$, which is a complete intersection. The Hilbert function of the quotient $\mathbb{C}[\partial]/F^\perp$ gives the lower bound matching (8); details in [8, 10].

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